

1. A rocket travels in the positive x -direction through force-free space. At some time, the rocket has mass m and velocity $\dot{x}$. The rocket ejects fuel backward with a constant exhaust speed v_{ex} relative to the rocket. During a short time interval, the rocket's mass changes by dm , where $dm < 0$, and its velocity changes by $d\dot{x}$. The positive mass of the expelled fuel is therefore $-dm$.

(a.) Write the total momentum immediately before and immediately after the fuel is ejected. Neglect products of small quantities, such as $dm d\dot{x}$, and use conservation of momentum to show that

$$P_i = m\dot{x}$$

$$m d\dot{x} = -v_{\text{ex}} dm.$$

$$P_f = (m + dm)(\dot{x} + d\dot{x}) + (-dm)(\dot{x} - v_{\text{ex}}), \quad dm d\dot{x} = 0$$

$$= m\dot{x} + m d\dot{x} + \cancel{\dot{x} dm} - \cancel{\dot{x} dm} + v_{\text{ex}} dm$$

$$P_i = P_f, \quad m\dot{x} = m\dot{x} + m d\dot{x} + v_{\text{ex}} dm$$

$$\therefore m d\dot{x} = -v_{\text{ex}} dm$$

b.) The rocket initially has mass m_0 and velocity $\dot{x}_0$. Separate and integrate the equation from part (a) to find the rocket's velocity $\dot{x}$ when its mass has decreased to m .

$$d\dot{x} = -v_{\text{ex}} \frac{dm}{m} \rightarrow \int_{\dot{x}_0}^{\dot{x}} d\dot{x}' = -v_{\text{ex}} \int_{m_0}^m \frac{dm'}{m'}$$

$$\dot{x} - \dot{x}_0 = -v_{\text{ex}} \ln\left(\frac{m}{m_0}\right)$$

$$\dot{x} = \dot{x}_0 + v_{\text{ex}} \ln\left(\frac{m_0}{m}\right)$$

Find the change in the rocket's velocity when its mass decreases from m_0 to $m_0/2$.

$$\Delta\dot{x} = v_{\text{ex}} \ln\left(\frac{m_0}{m_0/2}\right) = v_{\text{ex}} \ln 2$$

2. Consider two particles, each with mass m . At $t = 0$, their positions and velocities are

$$\mathbf{r}_1(0) = -a\hat{\mathbf{x}}, \quad \dot{\mathbf{r}}_1(0) = v_0\hat{\mathbf{y}},$$

$$\mathbf{r}_2(0) = a\hat{\mathbf{x}}, \quad \dot{\mathbf{r}}_2(0) = v_0\hat{\mathbf{x}}.$$

The external forces acting on the particles are

$$\mathbf{F}_1 = 0, \quad \mathbf{F}_2 = F_0\hat{\mathbf{x}},$$

where F_0 is a positive constant.

(a.) Find the initial position $\mathbf{R}(0)$ and initial velocity $\dot{\mathbf{R}}(0)$ of the center of mass.

$$M = m + m = 2m$$

$$\vec{R}(0) = \frac{\vec{r}_1(0)m + \vec{r}_2(0)m}{2m} = \frac{-a\hat{x} + a\hat{x}}{2} = 0\hat{x}$$

$$\dot{R}(0) = \frac{\dot{r}_1(0)m + \dot{r}_2(0)m}{2m} = \frac{v_0}{2}(\hat{x} + \hat{y})$$

(b.) Find the acceleration $\ddot{\mathbf{R}}(t)$ of the center of mass.

$$M\ddot{R} = \vec{F}_{\text{ext}} \rightarrow 2m\ddot{R} = F_0\hat{x}$$

$$\therefore \ddot{R} = \frac{F_0}{2m}\hat{x}$$

(c.) Find the velocity $\dot{\mathbf{R}}(t)$ and position $\mathbf{R}(t)$ of the center of mass.

$$\dot{R}(t) = \int \ddot{R}(t) dt = \dot{R}(0) + \frac{F_0 t}{2m} \hat{x} = \left(\frac{v_0}{2} + \frac{F_0}{2m}t\right)\hat{x} + \frac{v_0}{2}\hat{y}$$

$$R(t) = \int \dot{R}(t) dt = R(0) + \left(\frac{v_0}{2}t + \frac{F_0}{4m}t^2\right)\hat{x} + \frac{v_0}{2}t\hat{y}$$

$$\dot{R}(t) = \left(\frac{v_0}{2} + \frac{F_0}{2m}t\right)\hat{x} + \frac{v_0}{2}\hat{y}$$

$$R(t) = \left(\frac{v_0}{2}t + \frac{F_0}{4m}t^2\right)\hat{x} + \frac{v_0}{2}t\hat{y}$$